



Hibernate mappings

1. Need and advantages
2. Inheritance mapping
3. Association mapping
4. Component mapping

- Hibernate supports various mapping strategies to represent relationships between Java objects and database tables.
- Various types of mappings that are available are
 - Inheritance mapping
 - Association mapping(one to one, one to many and many to many)
 - Component mapping etc
- Programmers get freedom of designing the entities using object oriented principles
- This also avoids writing lot of joins for fetching data from mutiple tables.

- There are 4 diff ways in which inheritance hierarchy can be mapped into database tables.
- Mapped superclass
 - The parent classes can't be entities. Tables get created for sub classes by merging the properties of super class and super class to be annotated with @MappedSuperclass
- Single Table (table per hierarchy):
 - The union of all the properties from the inheritance hierarchy is mapped to one table
 - As all the data goes in one table, a discriminator is used to differentiate between different type of data.
 - Performance wise better
 - Many null values in the table

- Table Per Class.
 - Each concrete subclass has its own table containing both the subclass and the base class properties.
 - It is better to declare super class as abstract otherwise separate table gets created for super class with it's own attributes
 - Super class attributes are repeated in subclass mapped tables
- Table Per Sub Class(JOINED strategy).
 - All the classes are mapped to its own table.
 - It's highly normalized but performance is not good
 - Primary key column of super class is repeated in all subclass tables as primary key in the subclass table as well referred as foreign key for the super class table

- Entities can contain references to other entities.
- These associations are represented using foreign key relationships in the underlying tables
- Hibernate supports 4 types of associations:
 - One to One
 - One To Many
 - Many To One
 - Many To Many
- Direction of Associations: An association can be unidirectional or bidirectional.
- Cascade operations
 - Cascade in Hibernate refers to performing update, delete, save operation for even the associated entity.

- Component Mapping represents the has-a relationship.
- In this case a class is having a reference to another class as it's member variable.
- A single table is enough which holds the information of all the attributes of both container object and contained object.
- Contained object's class is annotated with `@Embeddable` and the property in the container class is annotated with `@Embedded`